

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**
**Perihalan Produk:**
**Product Description:**
**Cat No. :**
**Methyl sulfoxide**
**Methyl sulfoxide**

D128-1; D128-4; D128-500; D128-500LC; D128POP-19; D128POP-50; D128POP-200; D128POPB-200; D128RB-200; D128RS-19; D128RS-50; D128RS-200; XXD12818LI; NC1601910; NC1777239; NC1919033; D128SS-19; NC1949486; NC2240486; NC2961790

**Synonyms**

Dimethyl sulfoxide; DMSO

**CAS No**

67-68-5

**Molecular Formula**

C2 H6 O S

**Relevant identified uses of the substance or mixture and uses advised against**
**Recommended Use**

Laboratory chemicals.

**Uses advised against**

No Information available

**Company**

 Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
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 Selangor Darul Ehsan, Malaysia.  
 Main line: +60 3-5525 7888

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**Emergency Telephone Number**

 Tel: +03-5525 7888  
 CHEMTREC Malaysia **1-800-815-308** (Malay)  
 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**
**Label Elements**
**Hazard Statements**
**Precautionary Statements**
**Prevention**

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

**Response**

P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

**Storage**

P403 + P235 - Store in a well-ventilated place. Keep cool

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## Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

Combustible liquid

DMSO readily penetrates skin and may carry other dissolved chemicals into the body

Toxic to terrestrial vertebrates

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component          | CAS No  | Weight % |
|--------------------|---------|----------|
| Dimethyl sulfoxide | 67-68-5 | <=100    |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### General Advice

If symptoms persist, call a physician. Show this safety data sheet to the doctor in attendance.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.

#### Ingestion

Do NOT induce vomiting. Get medical attention.

#### Inhalation

Remove to fresh air. Get medical attention immediately if symptoms occur. If not breathing, give artificial respiration.

#### Self-Protection of the First Aider

No special precautions required.

### Most important symptoms and effects, both acute and delayed

Difficulty in breathing. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.

### Indication of any immediate medical attention and special treatment needed

#### Notes to Physician

Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### Suitable Extinguishing Media

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

#### Extinguishing media which must not be used for safety reasons

No information available.

### Special hazards arising from the substance or mixture

Combustible material. Containers may explode when heated. Thermal decomposition can lead to release of irritating gases and vapors.

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## Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Sulfur oxides, Sulfides, Formaldehyde.

## Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges. Ensure adequate ventilation.

### Environmental precautions

Should not be released into the environment. Do not flush into surface water or sanitary sewer system. See Section 12 for additional Ecological Information.

### Methods and Material for Containment and Cleaning Up

Remove all sources of ignition. Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Keep away from open flames, hot surfaces and sources of ignition. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation.

### Conditions for Safe Storage, Including any Incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component          | European Union | The United Kingdom | Germany                                                                                                                                                                                                                                                              |
|--------------------|----------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dimethyl sulfoxide |                |                    | TWA: 50 ppm (8 Stunden). AGW - exposure factor 2<br>TWA: 160 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 50 ppm (8 Stunden). MAK<br>TWA: 160 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 100 ppm<br>Höhepunkt: 320 mg/m <sup>3</sup><br>Haut |

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## Exposure Controls

### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

## Personal protective equipment

### **Eye Protection**

Wear safety glasses with side shields (or goggles)

### **Hand Protection**

Protective gloves

### **Skin and body protection**

Long sleeved clothing

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

### **Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

### **Recommended Filter type:**

Particle filter

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

## Environmental exposure controls

Prevent product from entering drains

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

#### **Appearance**

Colorless

#### **Physical State**

Liquid

#### **Odor**

Odorless

#### **Odor Threshold**

No data available

#### **pH**

No information available

#### **Melting Point/Range**

18.4 °C / 65.1 °F

#### **Softening Point**

No data available

#### **Boiling Point/Range**

189 °C / 372.2 °F

#### **Flash Point**

87 °C / 188.6 °F

**Method -** No information available

#### **Evaporation Rate**

No information available

#### **Flammability (solid,gas)**

Not applicable

Liquid

#### **Explosion Limits**

**Lower** 2.6 Vol%

**Upper** 42 Vol%

#### **Vapor Pressure**

0.55 mbar @ 20°C

#### **Vapor Density**

2.7

(Air = 1.0)

#### **Specific Gravity / Density**

1.100

#### **Bulk Density**

Not applicable

Liquid

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|                              |                          |
|------------------------------|--------------------------|
| Water Solubility             | Soluble                  |
| Solubility in other solvents | No information available |

## Partition Coefficient (n-octanol/water)

|                    |         |
|--------------------|---------|
| Component          | log Pow |
| Dimethyl sulfoxide | -1.35   |

|                          |                   |
|--------------------------|-------------------|
| Autoignition Temperature | 301 °C / 573.8 °F |
|--------------------------|-------------------|

|                           |         |
|---------------------------|---------|
| Decomposition Temperature | > 190°C |
|---------------------------|---------|

|           |                   |
|-----------|-------------------|
| Viscosity | 1.98 mPa.s @ 25°C |
|-----------|-------------------|

|                      |                                        |
|----------------------|----------------------------------------|
| Explosive Properties | explosive air/vapour mixtures possible |
|----------------------|----------------------------------------|

|                      |                          |
|----------------------|--------------------------|
| Oxidizing Properties | No information available |
|----------------------|--------------------------|

|                   |           |
|-------------------|-----------|
| Molecular Formula | C2 H6 O S |
|-------------------|-----------|

|                  |       |
|------------------|-------|
| Molecular Weight | 78.13 |
|------------------|-------|

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Hygroscopic.

### Possibility of Hazardous Reactions

|                          |
|--------------------------|
| Hazardous Polymerization |
| Hazardous Reactions      |

Hazardous polymerization does not occur.  
Thermal decomposition can take place above 189°C / 372°F.

### Conditions to Avoid

Incompatible products. Excess heat. Exposure to moist air or water. Keep away from open flames, hot surfaces and sources of ignition.

### Incompatible Materials

Strong oxidizing agents. Strong acids. Strong bases. Alkali metals.

### Hazardous Decomposition Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Sulfur oxides. Sulfides. Formaldehyde.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Product Information

(a) acute toxicity;

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| <b>Oral</b>        | Based on available data, the classification criteria are not met |                            |                              |
|--------------------|------------------------------------------------------------------|----------------------------|------------------------------|
| <b>Dermal</b>      | Based on available data, the classification criteria are not met |                            |                              |
| <b>Inhalation</b>  | Based on available data, the classification criteria are not met |                            |                              |
| Component          | LD50 Oral                                                        | LD50 Dermal                | LC50 Inhalation              |
| Dimethyl sulfoxide | LD50 = 28300 mg/kg ( Rat )                                       | LD50 = 40000 mg/kg ( Rat ) | LC50 > 5.33 mg/L ( Rat ) 4 h |

(b) skin corrosion/irritation; Based on available data, the classification criteria are not met

(c) serious eye damage/irritation; Based on available data, the classification criteria are not met

(d) respiratory or skin sensitization;  
Respiratory Based on available data, the classification criteria are not met  
Skin Based on available data, the classification criteria are not met

(e) germ cell mutagenicity; Based on available data, the classification criteria are not met

(f) carcinogenicity; Based on available data, the classification criteria are not met  
There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; Based on available data, the classification criteria are not met

(h) STOT-single exposure; Based on available data, the classification criteria are not met

(i) STOT-repeated exposure; Based on available data, the classification criteria are not met  
Target Organs None known.

(j) aspiration hazard; Based on available data, the classification criteria are not met

Symptoms / effects,both acute and delayed Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

Ecotoxicity effects Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants. Do not empty into drains. .

| Component          | Freshwater Fish                         | Water Flea         | Freshwater Algae               | Microtox                                                                                                                                                    |
|--------------------|-----------------------------------------|--------------------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dimethyl sulfoxide | 40 g/L LC50 96 h<br>33-37 g/L LC50 96 h | EC50 24h 7000 mg/L | EC50 96h 12350 -<br>25500 mg/L | = 16000 mg/L EC50<br>Pseudomonas putida 16<br>h<br>= 32 g/L EC50<br>Tetrahymena pyriformis<br>24 h<br>= 77 mg/L EC50<br>Photobacterium<br>phosphoreum 5 min |

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## Persistence and degradability

### **Persistence**

Persistence is unlikely.

### **Degradation in sewage treatment plant**

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants.

## Bioaccumulative potential

Bioaccumulation is unlikely

| Component          | log Pow | Bioconcentration factor (BCF) |
|--------------------|---------|-------------------------------|
| Dimethyl sulfoxide | -1.35   | No data available             |

## Mobility in soil

The product is water soluble, and may spread in water systems. . Will likely be mobile in the environment due to its water solubility. Highly mobile in soils.

## Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

## Other adverse effects

No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

#### **Waste from Residues/Unused Products**

Dispose of in accordance with local regulations

#### **Contaminated Packaging**

Empty containers should be taken to an approved waste handling site for recycling or disposal

#### **Other Information**

Do not flush to sewer

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

Not regulated

### Road and Rail Transport

Not regulated

### IATA

Not regulated

### **Special Precautions for User**

No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### **International Inventories**

X = listed

| Component          | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|--------------------|-----------|------|-----|-------|------|------|-------|------|----------|
| Dimethyl sulfoxide | 200-664-3 | X    | X   | X     | X    | X    | X     | X    | KE-32367 |

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## National Regulations

**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### **Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Revision Date**

24-Mar-2025

**Revision Summary**

Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

### **Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**